

Acoustics of Periodic and Multiple Drop Impacts on a Water Surface

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Abstract—High-speed video filming of surface currents and synchronized acoustic measurements of the underwater sound signals of falling drops were performed in a laboratory tank. During successive falling, the main structural elements of collision of a single drop with the surface are preserved in distorted form in the flow pattern: cavity, splashes, crown, and splash; shock pulses accompanying each contact are stably repeated in the phonogram. In addition, rare resonance packets are observed. For multiple falling drops, the flow pattern changes dramatically: the main structural elements of the drop impact flow disappear, and the surface is covered with floating bubbles. The phonogram assumes the form of a noise signal, in the spectrum of which separate linear sections stand out.

Keywords: drops, periodic and multiple impact, flow pattern, acoustics, spectra, scales, frequencies

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INTRODUCTION

Recent decades have witnessed significant expansion in studies on wave formation processes—initially capillary–gravitational [1], then acoustic [2, 3]—accompanying the merging of a free-falling drop with a target liquid. The growth in interest is due to the rich scientific content of the topic, an expanded number of practical applications, and the development of high-resolution technologies for the optical and acoustic recording of the parameters of physical processes in liquids. A search is underway for cause-and-effect relationships between simultaneously ongoing processes involved in the formation of groups of relatively low-frequency and small-scale capillary waves and high-frequency acoustic waves, the lengths of which noticeably exceed the dimensions of the source [4].

In early experiments [5], a rich composition of acoustic signals from the falling of a single drop had already been established, including a high-frequency primary pulse and a number of delayed resonance packets, the frequencies of which turned out to be close to the natural frequencies of oscillations of spherical gas bubbles [6]. The review of early works on the hydrodynamics and acoustics of the impact of a drop [7] is regularly supplemented with new results from studying the formation pattern of gas bubbles [8, 9] and acoustic signals [10–13], as well as data on the matched recording of the patterns of flows and sound fields [14, 15].

Careful studies have made it possible to single out the important role of relatively large gas bubbles

detached during splash submergence and a drop ejected from the top of this splash as one of the main sources of sound packets [15, 16]. Analysis of the results of [6–14] shows that the nature of the acoustic processes of a drop impact depends on a large number of parameters and is comparable in complexity with other phenomena studied in physical acoustics [17].

A number of key issues, including determination of the conditions for the emission of a tone [12, 13, 15] or modulated signal with a varying carrier frequency [11], require additional study in which merging pattern of the falling drop is controlled. In experiments, various modes of drop spreading with the formation of a cavity, crown, and splash (cumulative jet) [15, 18, 19] are observed; drop rebound with excitation of Rayleigh oscillations; suspension on the concave free surface of the liquid; and partial rebound with loss of some of the drop mass.

The flow mode depends on many parameters, such as the size, shape, and dynamic state of the drop, the intensity and mode of Rayleigh oscillations [21]; short capillary waves propagating along its surface [22], the orientation of the elongated drop, and the state of the surface of the target liquid, which can be smooth or perturbed by waves from external sources [23] or by previous drops, in particular, raindrops [24]. Differences in the properties of processes that depend on many heterogeneous parameters make it difficult to search for the formation mechanisms of gas bubbles and acoustic packets generated by them, depending on the experimental conditions for single falling drops or groups thereof [4, 8, 9, 11, 15, 19, 25].

The increased interest in studying the hydroacoustics of groups of drops and rain noise stems from a number of reasons. Even the first sea noise measurements [26] demonstrated the significant contribution of raindrop sounds to the sound field energy [27]. Further studies made it possible to identify the regional features of sound signals and occurrence of seasonal variability [28–30]. New results of sea measurements have resulted in laboratory experiment methods and study of the dynamics and energetics of acoustic radiation from single drops [20, 31] and the rain model [32].

The possibility of reconstructing rain parameters from remote measurement data on ocean noise [32] has become a rapidly growing scientific field in remote acoustic and radar meteorology [33]. At the same time, automated systems for measuring precipitation intensity have been created, making it possible to determine the size and energy of drops [34] and replacing labor-intensive manual measurements with a “precipitation bucket,” included in the standard equipment of meteorological stations [35]. Spectral analysis of marine remote sensing data focuses on identifying a spectral peak, the position and manifestation degree of which depend on the size of raindrops, wind speed, and precipitation intensity [36]. The acoustics of the intermediate range of the falling of a quantifiable number of drops remains insufficiently studied.

The objective of this study is to record and analyze sound signals in water caused by drops falling sequentially from one or simultaneously from a set of stationary dispensers with simultaneous optical recording of the flow pattern.

PARAMETERIZATION OF THE PROCESS

The experimental procedure is based on a complete system of fundamental equations describing the flows of liquids and gases with physically substantiated initial and boundary conditions [37, 38], for brevity not given here. The structure and dynamics of flows are determined by the properties of all contacting media, each of which is characterized by its own thermodynamic parameters indicated by subscripts: a (air); d (drops); t (target medium): density $\rho_{a,d,t}$, temperature $T_{a,d,t}$, coefficients of kinematic viscosity $\nu_{a,d,t}$ and surface tension σ_d^a and σ_t^a at the interfaces between media (values of $\gamma_d^a = \sigma_d^a/\rho_d$ and $\gamma_t^a = \sigma_t^a/\rho_t$, normalized to the density are also introduced). In this article, when studying flows and sounds accompanying the spreading in water of drops in free-fall with acceleration g , the influence of the air medium is neglected. The temperatures of all media are considered constant (room temperature) and unchanging due to the large heat capacity of water.

The flow parameters depend on the diameter D , surface area S_d , volume V , mass M , drop velocity U

at the moment of contact, kinetic energy $E_k = MU^2/2$, available potential surface energy (APSE) $E_\sigma = \sigma S_d$ concentrated in a near-surface spherical layer with a thickness on the order of the size of a molecular cluster $\delta_\sigma \sim 10^{-6}$ cm (see [39]). The characteristic time of kinetic energy transfer is determined by the drop spreading time $\tau_d \sim D/U \approx 2$ ms, and the rapid transformation of APSE into other forms is determined by the rate of breakdown of the free surface during the merging of liquids $\tau_c \sim \delta_\sigma/U \approx 10^{-8}$ s. The high density of surface energy concentrated in a thin layer ensures the formation of fast jets and drops at the boundary of the merging zone of contacting liquids [18].

Taking into account how fast processes of the transformation of internal energy into other forms greatly influence the dynamics and structure of flows, the Gibbs potential (free enthalpy) is chosen as the basic parameter describing the thermodynamic state of an medium in equilibrium G , the derivatives of which determine the main thermodynamic parameters: density, pressure, and temperature [40].

The problem is characterized by a large number of length and time scales, in particular, linear scales of a geometric (drop diameter D and dimension of the working volume L), kinematic ($L_g^U = U^2/g$), and physical nature. The length of capillary $\lambda_c = 2\pi/k$ or sound waves $\lambda_s = 2\pi/k_s$ is determined by their frequency and is given by the regular part of the proper dispersion relations. With allowance for viscosity, the dispersion relation for capillary waves has the form [41],

$$2k(\omega_c^2 k_l - gk^2 - \sigma k^4 + 2i\omega_c \nu k_l(3k^2 - k_l^2)) - (k_l^2 + k^2)(\omega_c^2 - gk - \sigma k^3 + 2i\omega_c \nu k^2) = 0, \quad (1)$$

and for sound waves [37],

$$\left(\mathbf{k}_s^2 \left(1 - \frac{i\omega_s \nu}{c_s^2} \right) - \frac{\omega_s^2}{c_s^2} \right) (\omega_s + i\nu \mathbf{k}_s^2)^2 = 0. \quad (2)$$

Dispersion equations (1), (2) describe the waves themselves and accompanying thin flows—ligaments with proper wavenumbers k_l [38, 41]. The system of equations also contains capillary–gravitational $L_g^\gamma = \sqrt{\gamma/g}$ and capillary–inertial scales $L_U^\gamma = \gamma/U^2$.

The group of small scales associated with the atomic and molecular properties of the medium includes the Prandtl scale $\delta_U^\nu = \nu/U$ and visco-capillary scale $\delta_\gamma^\nu = \nu^2/\gamma$.

The flow is also characterized by a set of time scales, one of which, $\tau_M = D/U$ is determined by the kinematics of the drop motion at the time of contact. The group of scales depending on the drop size and

Table 1. Characteristic values of dimensionless ratios in given experimental conditions

	$10^{-3} \times \text{Re}$	Fr	We	Bo	$10^3 \times \text{Oh}$	Ca
I: $D = 0.5$ cm, $U = 4.2$ m/s	21	356	1200	6.7	1.7	0.057
II: $D = 1.7\text{--}0.3$ cm, $U = 2.8$ m/s	4.8–8.5	476–270	185–326	2.3–4.0	2.8–2.1	0.038

medium properties includes $\tau_\gamma = \sqrt{D^3/\gamma}$ and $\tau_\gamma^D = \nu D/\gamma$. Two more pairs of scales depend on the parameters of the medium $\tau_g^\gamma = (\gamma/g^3)^{1/4}$, $\tau_\gamma^\nu = \nu^3/\gamma^2$ and the source of acoustic $\tau_s = 2\pi/\omega_s$ and capillary waves $\tau_c = 2\pi/\omega_c$.

The scales of the studied phenomena span a wide range: linear, from 160 to 10^{-6} cm; time, from 10^{-2} to 10^{-8} s. The upper limit of the linear scales is typical of mechanical processes, while the lower limit is close to the atomic–molecular scales (the size of water molecules is 3×10^{-8} cm, the cluster size is about 10^{-6} cm, see [39]).

In general, the sets of scales determine the requirements for experimental setups: the choice of the size of the observation field, which should contain all large-scale components (in particular, the propagation regions of capillary waves with a length on the order of δ_ν^γ) and tools to resolve rapidly changing thin flows.

The large ranges of the scales proper indicate the coexistence of two types of influencing processes. One group of scales characterizes slow hydrodynamic phenomena ($\sim \tau_M$), and the second, fast thin ones, at the boundary of atomic–molecular and macroscopic processes. Their joint action is clearly manifested in the evolution of the merging pattern of independent free surfaces, when the APSE is converted into other forms (internal and mechanical energy) as a result of direct atomic–molecular interactions and a sharp change in thermodynamic quantities (pressure and temperature) occurs in a thin layer. The rapid release of energy contributes to the formation of an initial expanding sheet, spikes, and splashes from the contact area [42, 43], drops from the points of the crown, and a splash [20], and high acceleration of the boundaries of detached gas cavities [11]. The scale ratios define dimensionless combinations, which were introduced separately in earlier works.

Following the established tradition, when describing the processes, linear scale ratios are used here, which specify the Reynolds $\text{Re} = D/\delta_\nu^\nu = UD/\nu$, Froude $\text{Fr} = \delta_g^U/D = U^2/gD$, Weber $\text{We} = D/\delta_\nu^\gamma = U^2D/\gamma$, Bond $\text{Bo} = D^2/\delta_g^\gamma = gD^2/\gamma$, and Ohnesorge $\text{Oh} = \sqrt{\delta_\nu^\nu/D} = \nu/\sqrt{\gamma D}$ numbers and the capillary ratio $\text{Ca} = U/U_\gamma^\gamma = U\nu/\gamma$. Table 1 presents the charac-

teristic values of the dimensionless ratios under the conditions of the present experiments.

EXPERIMENTAL TECHNIQUE

The experiments were carried out on a installation at the Ishlinsky Institute for Problems in Mechanics RAS [44]. The working tank was $240 \times 40 \times 60$ cm in size with optical glass and was filled with partially degassed tap water. Experiments were carried out at room temperature $T = 20^\circ\text{C}$.

Video was shot on an Optronis 3000 $\times 2$ camera at a frame rate of 4000 fps, frame size of 512×512 , and an absolute spatial resolution of 0.07 mm. The axis of the video camera was directed at an angle of 70° to the surface.

Sound signals were recorded by a GI54 hydrophone with a preamplifier (frequency band 0.002–100 kHz, ripple 3 dB, output slope 30 mV/Pa). The hydrophone was placed in the center of the tank at a depth of 5 cm, on the line of incidence of a single drop, or at the geometric center of incidence of a group of drops. This arrangement made it possible to reduce the masking effect of reverberation associated with scattering from the bottom and walls of the tank. The sampling frequency of the hydrophone signal is up to 1 MHz. The technique is discussed in more detail in [11, 15].

Two sources of drops were used in the experiments. In experiment I (Table 1), a single drop dispenser with a 0.4 cm capillary was used, which made it possible to control the period of their detachment from several tens of seconds to 10 ms; it was located at a height of 110 cm.

In experiment II, a multiple drop dispenser was used, consisting of 80 capillaries with a diameter $d_c = 0.5$ mm spaced evenly in concentric circles with a maximum diameter of 6.0 cm. The source was located at a height of 50 cm. The drop sizes from the multiple source ranged from 1.7 to 3.0 mm at an average contact rate with the liquid surface of $U = 2.8$ m/s.

HYDRODYNAMICS AND ACOUSTICS OF THE IMPACT OF PERIODICALLY FALLING DROPS FROM A FIXED DISPENSER (EXPERIMENT I)

In a series of these experiments, drops with a diameter $D = 0.5$ cm, falling with a period of 220 to 600 ms, reached a speed of $U = 4.2$ m/s at the moment of con-

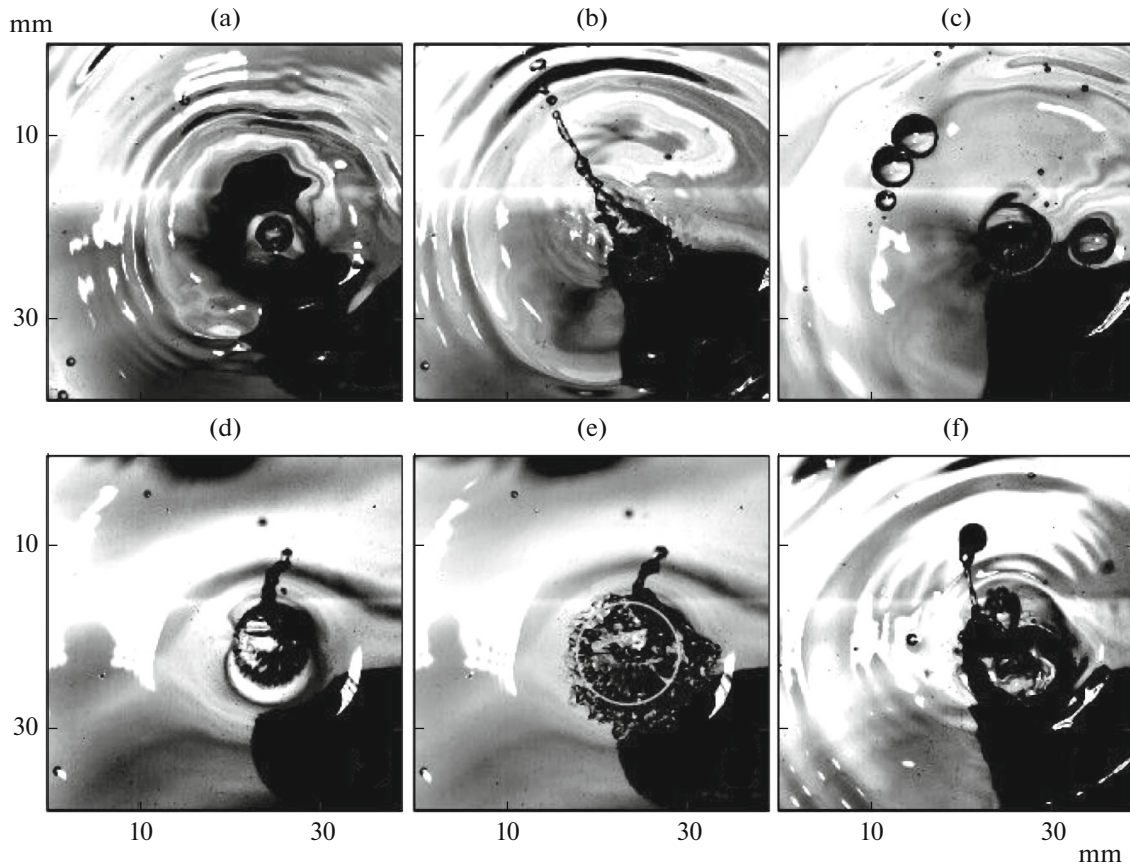


Fig. 1. Photographs of surface perturbations for periodic ($T = 218$ ms) falling of drop: (a–f) $t = 229, 251, 1317, 2512, 2515, 3002$ ms. Light circle (e) is scanning trace of illumination profile.

tact with the unperturbed surface (measured from video films). The dimensionless parameters of the experiment are given in Table 1.

The sequence of video frames shown in Fig. 1 is selected from a video time series with an average period of 220 ms.

Analysis of the video films shows that the recorded flow patterns retain all the basic structural components that arise when a single drop merges with a free surface: small splashes of the primary contact [42, 43]; a cavity; a crown, along the surface of which capillary waves run down [45] and from spikes on the tops of the teeth groups of drops are emitted; a central splash; secondary caverns; and a thin central jet—a streamer [46].

The appearance and decay of each structural element is accompanied by a group of capillary waves. However, the shapes of the structural elements are far from the regular ones observed in the falling of a single drop [20]. Thus, in this sampling (Fig. 1a), the phase surfaces of annular capillary waves are curved closed lines, deviation of which from a circle increases closer to the center. The average wavelength in the direction 0–10 o'clock is $\lambda_c = 2.2$ mm.

The thin central jet (streamer) that deviated from the vertical is more extended and contains a significantly larger number of drops (Fig. 1b) than in the liquid at rest. A peculiarity of the flow is a large number of floating gas bubbles of various diameters, from 0.2 to 0.7 cm (Fig. 1c). When a drop merges with a perturbed surface, the fine-structure components deforming the top of the growing splash are more pronounced at the periphery of the flow. Some are so intense that they generate their own off-center jet (Fig. 3d). Within 3 ms, small-scale perturbations cover the entire central region. In the spatial spectrum of fluctuations of the illumination along a circular contour with a radius of 0.5 cm (Fig. 1e), they stand out on a scale of 1.8 mm along with large-scale perturbations of a scale of 8.8 mm. Subsequent calming of perturbations is accompanied by a group of capillary waves ($\lambda_c = 2.4$ cm) in the direction of one o'clock (Fig. 1f). All the figures show floating and near-surface gas bubbles.

Two groups of signals are distinguished in the hydrophone phonogram (Fig. 2a): shock pulses I and resonance packets II and III. A shock pulse arises at the contact of each falling drop; its structure is analyzed in detail in [46, 47].

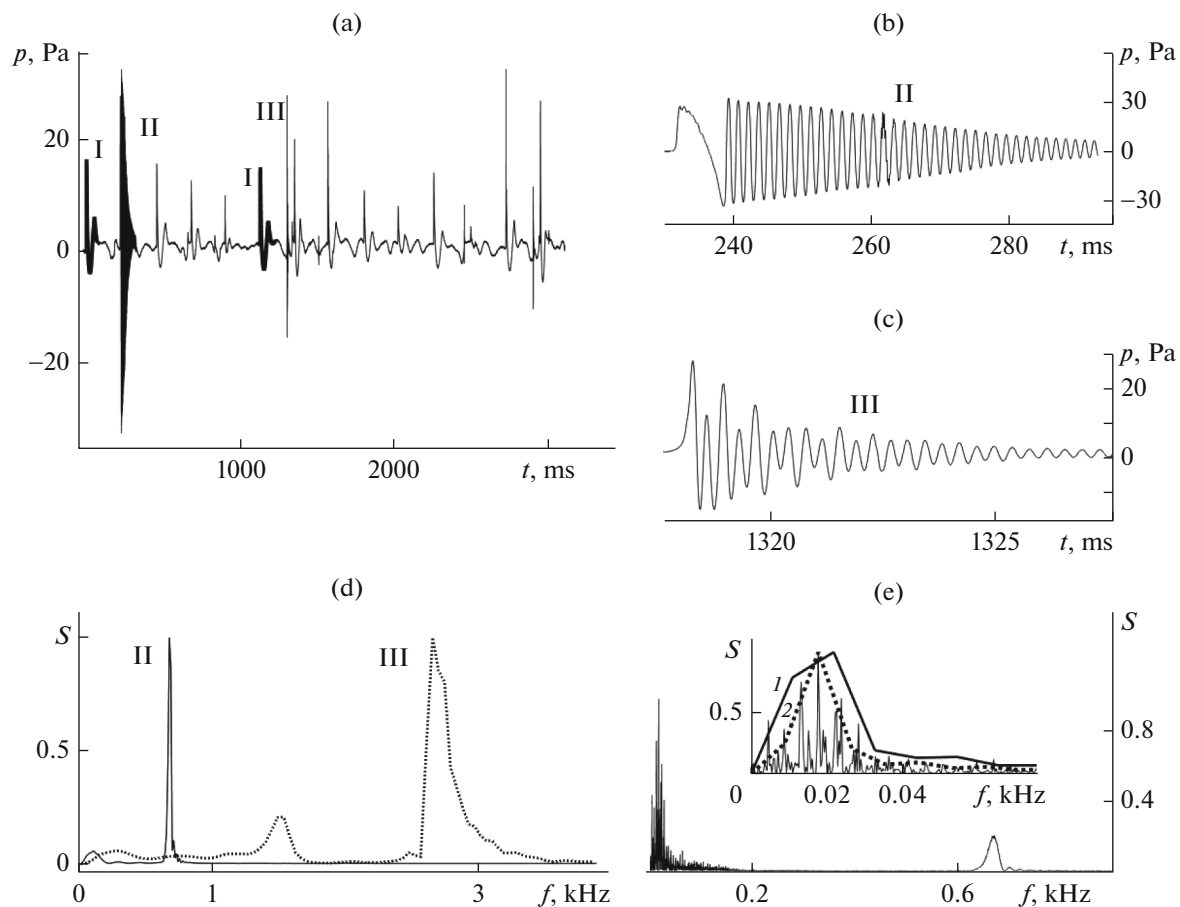


Fig. 2. (a) Total phonogram with periodic ($T = 218$ ms) falling of drop; I, shock pulse; II, III, resonance packets and (b, c) their expansions; (d) amplitude spectra of resonant packets II and III; (e) the amplitude spectrum of the complete phonogram, the inset is the expanded initial segment, curves 1 and 2 are the amplitude spectra of shock pulses I (highlighted in bold in Fig. 2a).

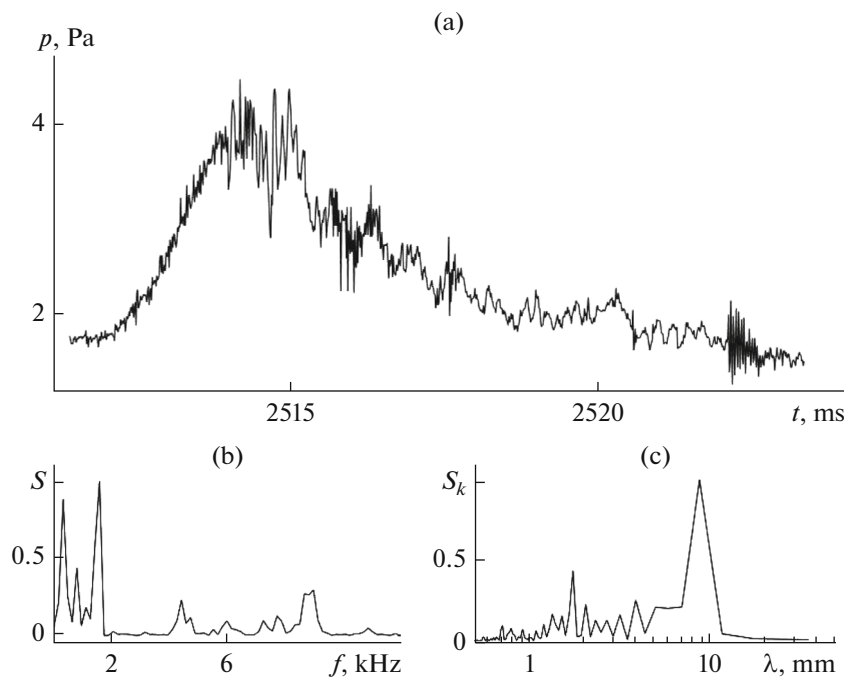


Fig. 3. (a) Expanded area at $t = 2515$ ms and (b) its frequency spectrum; (c) spatial spectrum of fluctuations in illumination along circular contour in Fig. 1d.

Table 2. Frequencies of experimentally determined peaks in area of line spectrum

	f_1		f_2	f_3		f_4		f_5			
Hz	4.48	7.04	8.64	13.12	15.04	17.6	18.88	22.4	23.68	28.16	32.64
n		1.57	1.93	2.92	3.36	3.92	4.21	5	5.28	6.28	7.28

The appearance of resonance packets is largely determined by the splashdown scenario, which is not reproduced under these conditions due to the variability in the shape of the free surface. In the present experiment, two resonance packets are distinguished, which differ in shape (Fig. 2b) and spectrum (Fig. 2d). Note that the amplitudes of the resonance packets exceed the shock pulses. Packet II is a damped monochromatic signal (Fig. 2b) with a frequency of 665 Hz, which corresponds to an oscillating air bubble with a diameter of 1.0 cm. The spectrum of the amplitude-modulated signal of packet III contains two peaks at frequencies of 1.48 and 2.65 kHz (Fig. 2d).

In the spectrum of the complete phonogram (Fig. 2e), two sections are distinguished: the initial line and a single peak at a frequency of 667 Hz—the resonance of a single bubble, also highlighted in the spectrum in Fig. 2d.

In the region of the linear spectrum, the lower cut-off frequency of 4.48 Hz corresponds to the drop falling frequency. Its harmonics with numbers $n = 2-5$ (f_2-f_5 , Table 2) and another six frequencies of unestablished nature are pronounced in the spectrum.

In the swept area of the phonogram (Fig. 3a), corresponding to the flow pattern with a pronounced fine structure (Fig. 1e), individual sound packets with an average duration of 100 μ s are visible in the descending section.

The frequency spectrum (Fig. 3b) contains peaks at $f = 314, 1570, 4400$, and 8960 Hz. In the spatial spectrum of surface perturbations of the liquid (light contour in Fig. 1e), a scale of 1.8 mm is distinguished, which reflects the microstructure formed by smooth troughs and sharp crests, as well as a much larger scale of 8.8 mm (Fig. 3c), formed by clusters of small inhomogeneities.

FLOW PATTERN AND ACOUSTICS OF THE IMPACT OF MULTIPLE DROPS (EXPERIMENT II)

Simultaneously falling drops complicate the flow pattern, which is dominated by floating bubbles and zones of darkening in places where drops fall and lightening in places where the surface is smoothed (Fig. 4). In a typical flow pattern, floating air bubbles occupy a large surface area.

The location and shape of the darkened areas change significantly from frame to frame (Figs. 4a–4d). In the first two frames, rounded dark spots are

pronounced in the central region, then a long sleeve appears (Fig. 4c); Fig. 4d shows a complex figure. Bubbles of various sizes accumulate in groups, the configuration of which changes rapidly.

A typical phonogram from a hydrophone installed in the center of the drop falling zone at a depth of 5 cm (Fig. 4e) on this time scale looks like a noise track with pick-to-pick of 5 Pa and individual burst up to 10 Pa. The spectra measured at times corresponding to the frames in Figs. 4a–4c start at a frequency of 1 kHz (Fig. 4f1–4f3), but their behavior differs significantly with increasing frequency. The spectrum in Fig. 4f1 has a wide bicuspid peak near 1.1 kHz and a small local maximum at 2.7 kHz. The spectrum in Fig. 4f2 occupies the same frequency range, but it is purely linear and spans a number of frequencies: 1.1, 1.2, 1.5, 1.7 and 2.0 kHz. In the spectrum in Fig. 4f3, there are only two single peaks at 1.1 and 2.7 kHz.

The spectrum of the total signal is dominated by low-frequency components in the $f < 50$ Hz range. All high-frequency components ($f > 1$ kHz) also presented in the spectra in Figs. 4f have significantly smaller amplitudes. They are shown in short groups near small peaks at frequencies of 1.1 and 2.7 kHz.

An extended fragment of a phonogram with a duration of 10 ms ($t = 1328-1338$ ms) illustrates the behavior of the acoustic signal near mark 1 (Fig. 4e). The initial section is low-frequency noise, which then turns into high-frequency oscillations, in the spectrum of which are two pronounced peaks at frequencies of 5.5 and 11.0 kHz. It should be noted that in earlier works in experiments on measuring the acoustic noise of artificial and natural [27, 48] rain, spectral peaks were observed in the range up to 15 kHz.

The high-frequency section of the total spectrum (Fig. 5a) contains peaks at frequencies that are unevenly scattered in the range of 1–8 kHz. The most significant peaks refer to frequencies of 0.1, 1.0, 2.7, 3.7, 5.5 kHz. In the stretched area (Fig. 5b) near mark 1 in Fig. 5a, individual lines are seen, the gap between which remains with further stretching of the frequency scale (Fig. 5c). A similar behavior is shown by area 2 in Fig. 5a: its stretched area (Fig. 5d) also contains separating local maxima on both sides of the main peak (7.8 kHz), which appear even more distinct in its stretched fragment (Fig. 5e).

Figure 6 shows the spatial spectra of fluctuations of the illumination of the optical pattern (Fig. 4a) taken along the diagonals (1, 2) and circles (3, 4) (Fig. 6a).

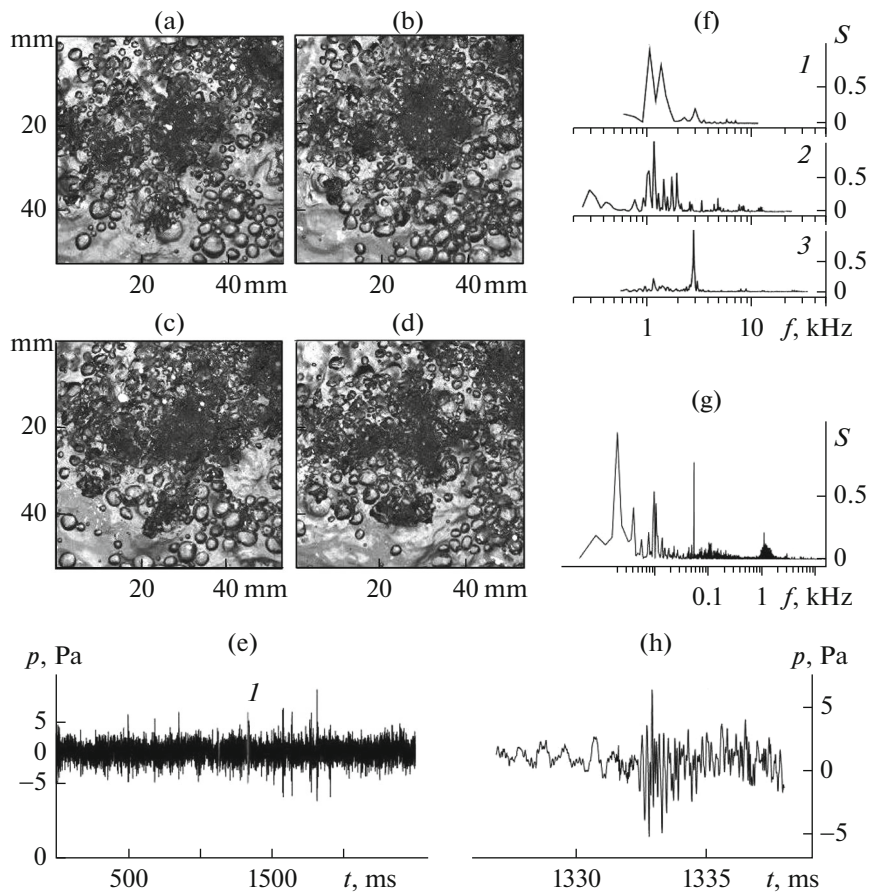


Fig. 4. (a–d) View of surface for falling of multiple drops, $t_c = 30, 472, 682, 1129$ ms; (e) total phonogram of artificial rain; (f1–f3) spectra of areas with duration of 10 ms for $t = t_c$; (g) spectrum, (h) expansion of area I of total phonogram.

Scanning of a video frame in Fig. 6a along the diagonals I and 2 gives the spatial spectra (Fig. 6b, curves I and 2) occupying the same scale ranges (0.3–25 mm) and containing a countable number of peaks. The peak frequencies of spectra I and 2 do not coincide, which is particularly pronounced in the small-scale region ($\lambda < 6.3$ mm) and less pronounced in the large scale region. The spectrum along the diagonal I is more rarefied and dominated by peaks near scales of 1.5, 3, 5, and 10 mm. The spectrum along the diagonal 2 is characterized by the frequent location of peaks with increasing amplitude in the range of 1.5–5 mm and decreasing in the range of 7–12 mm. Small scales correspond to bubble sizes in the lower and upper corners of the diagonal; large scales correspond to dark clusters near the center—points of drop contact.

The distribution of scales along curvilinear trajectories—in this case along circles 3 ($R = 10$ mm) and 4 ($R = 22$ mm)—looks more discrete, especially in the small-scale part; at the same time, it shows a strong difference in the composition of the spectral lines and envelope (Fig. 6b, curves 3 and 4). The spectrum along small circle 3 contains two peaks ($\lambda = 4.8$ and

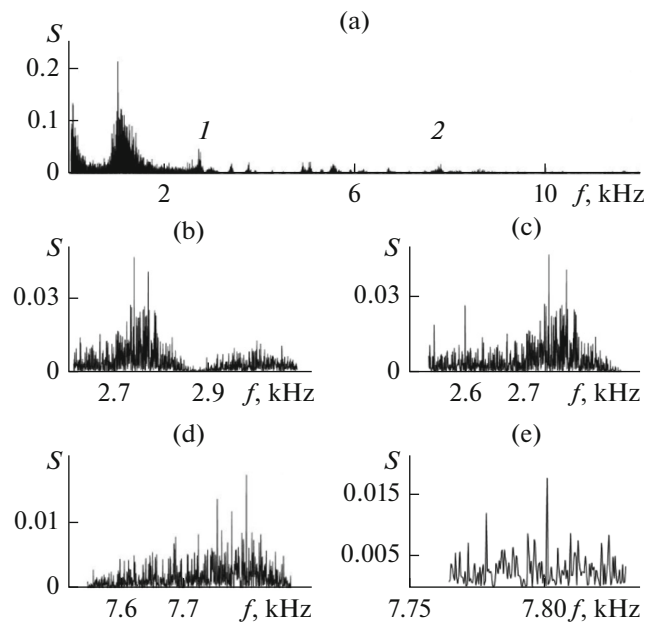


Fig. 5. (a) Expansion of high-frequency part of spectrum of total phonogram (Fig. 4g), (b–e) its stretched fragments in vicinity of local maxima.

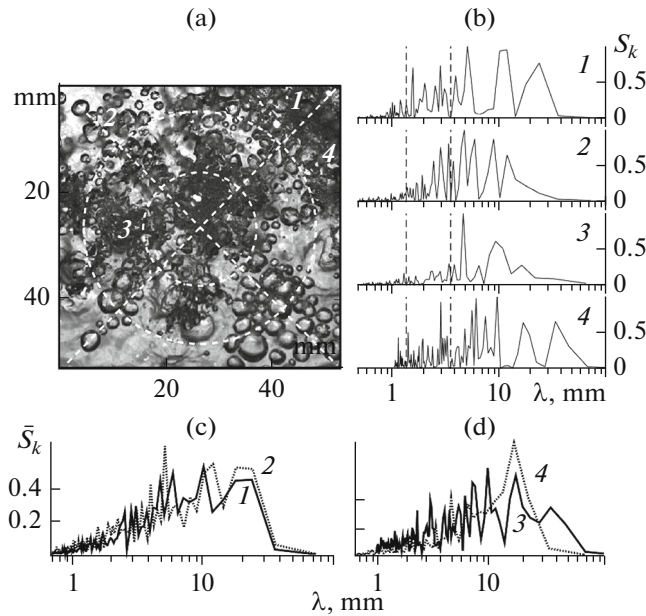


Fig. 6. (a) Scanning trajectories of illumination at $t = 30$ ms; (b) normalized spectra S_k of fluctuations in illumination along diagonals 1, 2 and circles 3, 4 at $t = 30$ ms (curves 1–4, range of measured sizes of falling drops is marked); (c, d) cumulative spectra at times $t = 30, 472, 500, 682, 1129$, and 2114 ms.

9.6 mm), which, taking into account the twofold difference in amplitudes, can be interpreted as the main (dominant) scale and subharmonic. The spectrum of large circle 4 differs sharply in the small-scale range by pronounced peaks owing to bubbles of different sizes randomly distributed along the circumference. The amplitudes of the maxima are comparable to the main maxima on scales $\lambda = 6.3$ and 10 mm. The largest peaks at $\lambda = 17.3$ and 34.6 mm are multiples and have the same amplitudes, which may indicate two periodic structures (clusters) with the given scales.

Figures 6c and 6d show the cumulative spectra obtained by averaging the spatial spectra over a number of time moments $t = 30, 472, 500, 682, 1129$, and 2114 ms. The cumulative spectra along the diagonals (Fig. 6c, 1 and 2) have similar features: they have the most distinct scales near 3 and 5 mm, which are also present at time $t = 30$ (Fig. 6b, 1 and 2).

Spectrum along the great circle (Fig. 6a, 4) except for the large-scale peak has only one pronounced maximum on a scale of 5 mm (Fig. 6d, 4), which is also present in the spectrum along the small circle (Fig. 6d, 3). The latter is almost linear and contains all the sizes of bubbles found in the pattern. The significant range of scales of surface perturbations is limited from below by the maximum size of falling drops (3 mm).

CONCLUSIONS

For the first time, experiments were carried out in which acoustic and optical images of perturbations created by periodic and multiple collisions of drops with a perturbed water surface have been obtained.

During periodic impact, the flow pattern retains the same structural elements (cavern, crown, splash, streamer) as in the case of a single drop; however, they lose their symmetry. After several collisions, floating bubbles with a diameter of 1 to 6 mm appear on the surface, which gradually increase in number.

The phonogram contains the same set of signals as during the impact of a single drop. In this case, a shock pulse accompanies each drop impact, while a resonance packet is only noted infrequently. The spectrum of the total phonogram has two pronounced maxima: a low-frequency one near 0.2 kHz (associated with the drop falling period) and a higher-frequency one at 0.7 kHz, due to the resonance sound of infrequent bubbles.

In the case of multiple impacts, the pattern of flows characterized by a single drop is completely absent. Groups of bubbles of various shapes and sizes float on the surface. The current spatial spectra appear linear, but the composition and location of the peak scales change at each time instant. In the cumulative spectra taken along straight lines and circles, linear segments also dominate, in which peaks are distinguished at frequencies that occur in the spectra for individual time instants.

The phonogram of multiple falling drops has contrasting sections, some of which have the form of a noise track, and others are individual resonance packets. Accordingly, the spectra of individual samplings are continuous, while others are a limited set of harmonics.

In the spectrum of the total phonogram, two areas are distinguished: low-frequency (up to 50 Hz), in which the main energy of the signal is concentrated, and high-frequency with a maximum cutoff frequency of 2.7 kHz.

For the practical implementation of laboratory data, it is advisable to increase the number of dispensers while simultaneously broadening the range of impact velocities.

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